CPSC 100
Computational Thinking

Human Computer Interaction

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Agenda

- Course Admin
  – Milestone 3
- Learning Goals
- Human Computer Interaction
  – Introduction + Activity

Course Admin

Project Milestone 3

Project Milestone 3

Research and Inquiry (Milestone 3)

This milestone involves a detailed presentation of the articles they have summarized and data they have found related to their specific questions intended for fellow students in the CPSC 100 class...

Students will:

1. Delve into their research, summarizing the findings and examining how they address the questions they were interested in.
2. Create an outline for a digital artifact that will visually represent their findings.

Project Milestone 3

- Due date: Wednesday March 4, 2026
- Open on PrairieLearn later this week

Milestone 3 Details

- This is your opportunity to demonstrate your knowledge and understanding.

- From the Project Contract Grade table, these items are the focus:
  - "Topics Covered"
  - "Complexity & Sophistication"

Milestone 3 Details

- **Each student** will write a short paper detailing their research of their area of interest using the course topics for their area of interest.
  - C: ~800 words (each student)
  - B: ~1000 words (each student)
  - A: ~1200 words (each student)
  - A+: ~1600 words (each student)

Milestone 3 Example

Milestone 3 Details

- The goal of the paper is to help you consolidate your ideas, your research, and your understanding of the topics in greater detail.
- This is where the "Topics Covered" and "Complexity & Sophistication" part of your term project comes in.

Milestone 3 Details

- The paper needs to be “well-sourced” ; you can choose any referencing method you like and are used to (APA, MLA, etc…)

- **Each course topic** you have in your project should have a **minimum of 3** sources (more is fine)

Milestone 3 Details

- The paper can be in “draft” form, but should be clearly written with mostly full sentences, free from typos, spelling mistakes.

- This paper will be the foundation of your ‘digital artifact’.

Milestone 3 Details

- The paper should be submitted as a PDF file on PrairieLearn and should include the word count clearly listed under the title.

- Note that the word count does not include your references/works cited

Contracted Grade Requirements

All requirements listed below are per group member !				
	C	B	A	A+
Topics Covered	• 4 course topics	• 5 course topics	• 6 course topics	• 8 course topics
Aesthetics & Visualizations	• No relevant and substantial visualizations present • No effort made on aesthetics	• Minimal relevant and substantial visualizations present • Minimal effort made on aesthetics	• Several relevant and subsantial visualizations • Significant effort made on aesthetics	• Originally-created and/or extremely high quality sourced visualizations • Significant effort made on aesthetics
Complexity & Sophistication	• Simple research questions	• Sophisticated research questions	• Deep and complex research questions • Analysis and commentary	• Deep and complex research questions • Deep analysis and insightful commentary • Extra flair and effort
Cohesiveness	• Separate reports for students • Mini projects for each student	• Single report with disparate topics • Minor spelling and proofreading	• Single report with all topics closely related	• Single report with all topics closely related • Coordinated styling
Effort & Group Dynamics	• Imbalanced workload distribution	• Reasonable workload distribution	• Equitable workload distribution	• Equitable workload distribution

CPSC 100 Course Project

CPSC 100

Computational Thinking

Intro to Human Computer Interaction

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Learning Goals

Learning Goals

After this **today's lecture**, you should be able to:

- Describe the **historical evolution** of HCI, highlighting pioneers like Douglas Engelbart and key innovations
- Describe the concept of **IoT** and give concrete examples (e.g., smart thermostats, wearable health devices).
- Distinguish between **AR** and **VR** technologies, and identify key examples (e.g., Google Glass, Meta Quest)
- Define and explain the **five key usability attributes**: learnability, efficiency, memorability, errors, and satisfaction.

Human Computer Interaction

Introduction to HCI

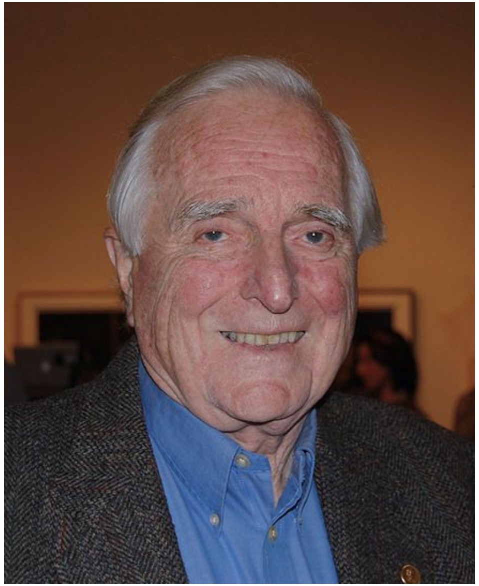
- Human-Computer Interaction (HCI) is the study and practice of **how people interact with computers** and design technologies that let humans engage with digital systems effectively and intuitively.

Where did it start from?

Douglas Engelbart

Douglas Engelbart (1925-2013)

- Founding father of HCI (one of)
- Augmentation Research Center
  - SRI International (Non-profit R&D org)
- Inventions
  - Computer mouse (1968)
  - NLS (oN-Line System - 1960s)



Douglas Engelbart: 2008

Reference: Ivan [Link]

What is challenging us now?

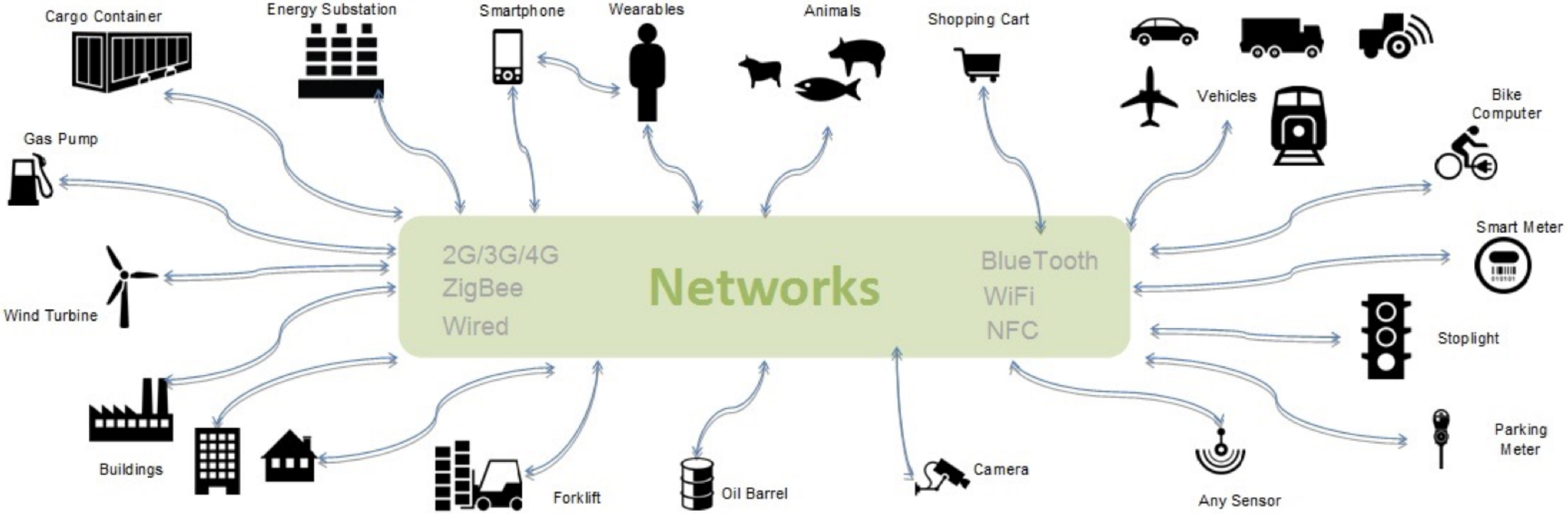

SMART TRACKER
SMART REFRIGERATOR
ASSISTANT SPEAKER
ROBOT CLEANER
SMART KETTLE
SMARTHOME SENSOR
SMART SCALE
COFFEEMACHINE
SMART WASHER
CCTV & IP CAMERA
SMART MICROWAVE
SMART HEATING PRESSURE
SMART SOCKET
SMART STOVE
InterviewBit


ORDER FOOD ONLINE
ARTIFICIAL INTELLIGENCE
Can I Help?

IoT

Internet of Things (IoT)

"**Things**" refer to any physical object with a device that has its **own IP address** and can **connect & send/receive** data via a **network**



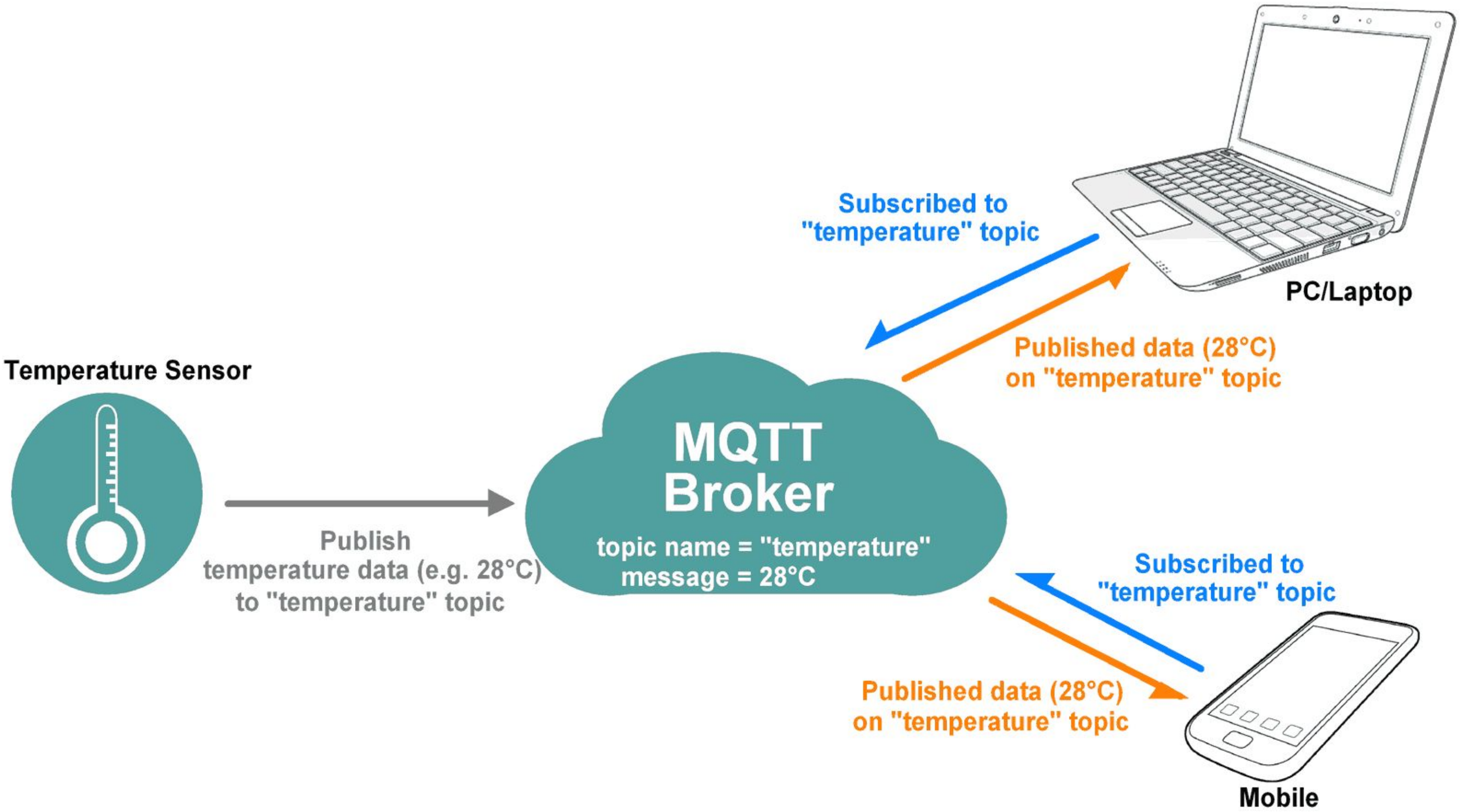
Reference: IBM [Link]

Internet of Things (IoT)

Everyday objects with connectivity, sensing abilities, and increased + embedded computing power.

- Connected home technology
  - Thermostats, lighting, energy monitoring
- Wearables
  - Activity/fitness trackers
- Medical/wellness devices
  - Bathroom scales, blood pressure monitors

What happens to IoT devices when there is no internet?


Subscribed to
"temperature" topic
PC/Laptop
Published data (28°C)
on "temperature" topic
Temperature Sensor
MQTT
Broker
topic name = "temperature"
message = 28°C
Publish
temperature data (e.g. 28°C)
to "temperature" topic
Subscribed to
"temperature" topic
Published data (28°C)
on "temperature" topic
Mobile

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AR + VR

Augmented + Virtual Reality

Virtual Reality (VR)

- Use of computers to simulate a real or imagined environment
- Three-dimensional (3-D) space

Augmented Reality (AR)

- Uses an image of an actual place or things that adds digital information to it

Meta Quest 2019-now

Google Glass 2014-15

Augmented + Virtual Reality

Participation Question

How do you feel about Meta's AR glasses?

Net positive for society	Net negative for society
	It's complicated…

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How do we *design* for the future?

HCI: User Centered Design

HCI: Usability

- Quality attribute
  - Assesses how easy user interfaces are to use
  - Improving ease-of-use during the design process
- Defined by 5 quality components

HCI: Usability Components

1. Learnability
2. Efficiency
3. Memorability
4. Errors
5. Satisfaction

HCI: Usability Components

- **Learnability:**
  - How easy is it to learn task the first time?

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Activity

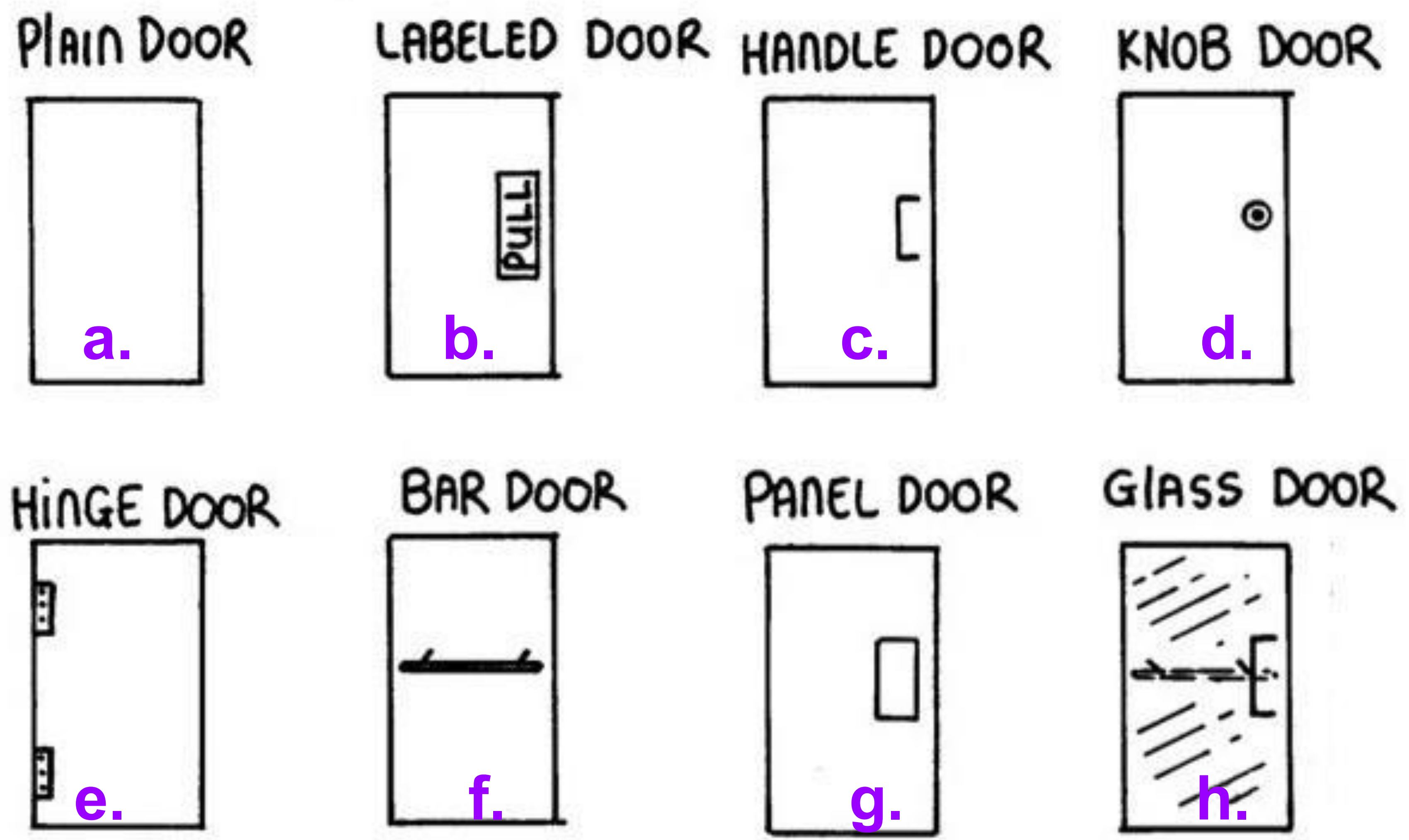
Intro to HCI Activity

- **Answer on PrairieLearn Class Activity for today**

  1. Think of a technological interaction from last week that irritated you.

  2. Draw/visualize it (to the best of your ability)

  3. Explain exactly HOW it failed for you. Depict activity, tasks, interactions.


PLAIN DOOR
LABELED DOOR
PULL
HANDLE DOOR
KNOB DOOR
a.
b.
c.
d.
HINGE DOOR
BAR DOOR
PANEL DOOR
GLASS DOOR
e.
f.
g.
h.

Q: How does this door work?

iClicker

A. Push to the left

B. Push to the right

C. Pull on the left

D. Pull on the right

E. Slide it along

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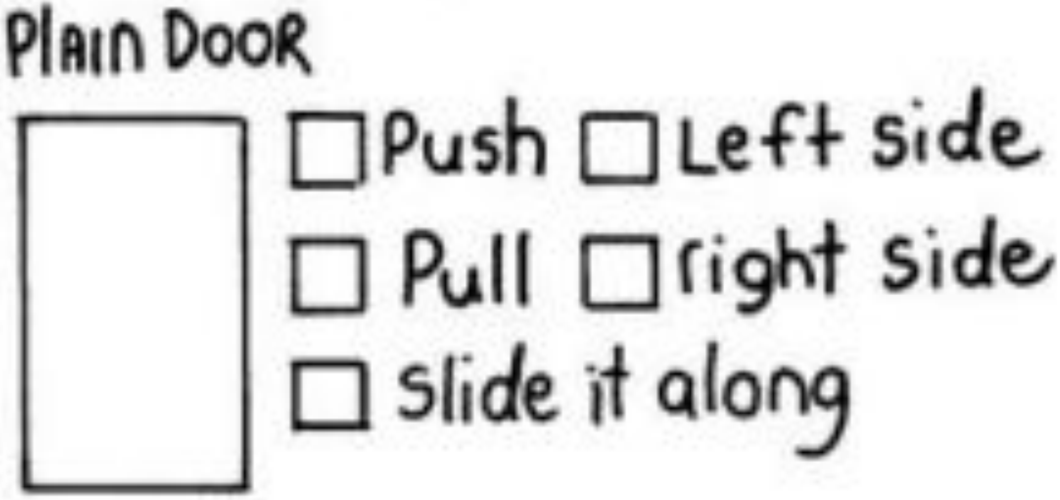

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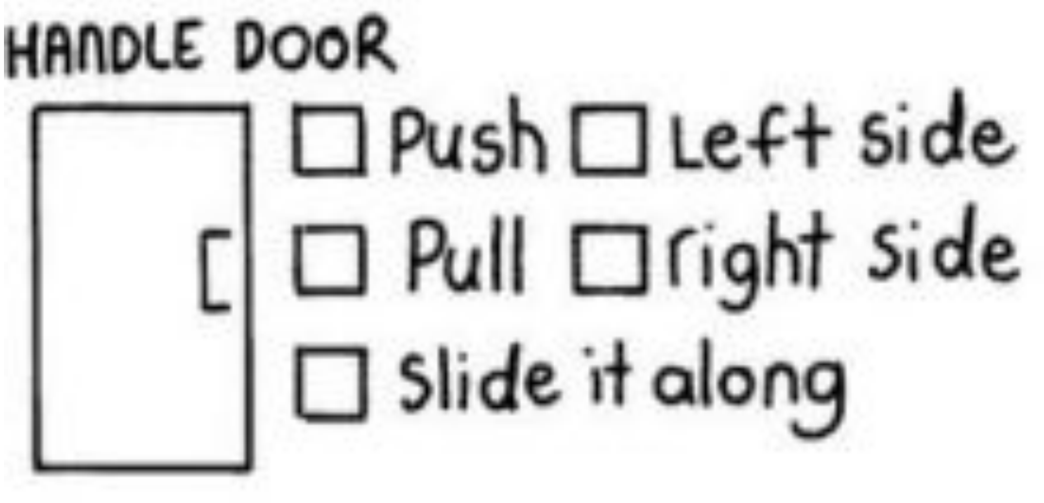

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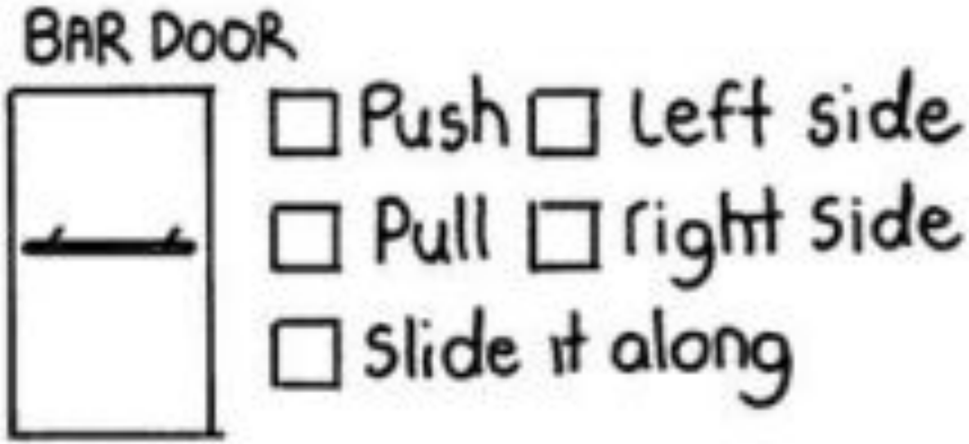

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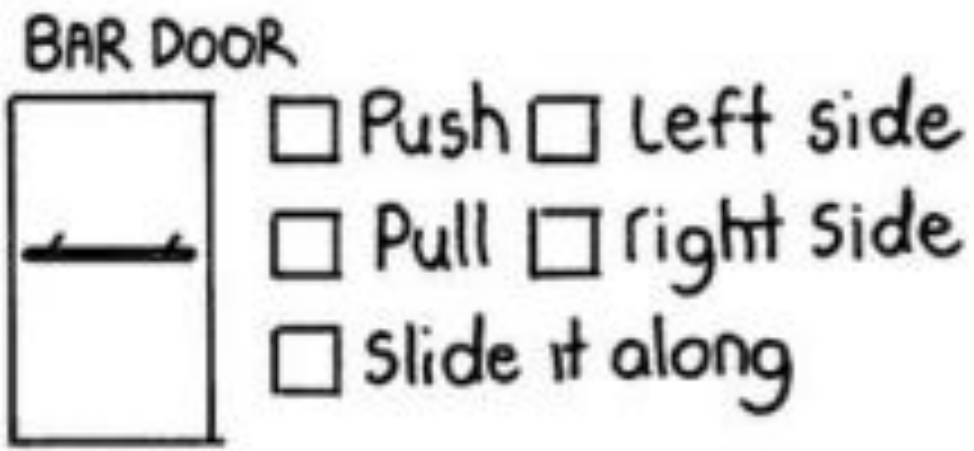
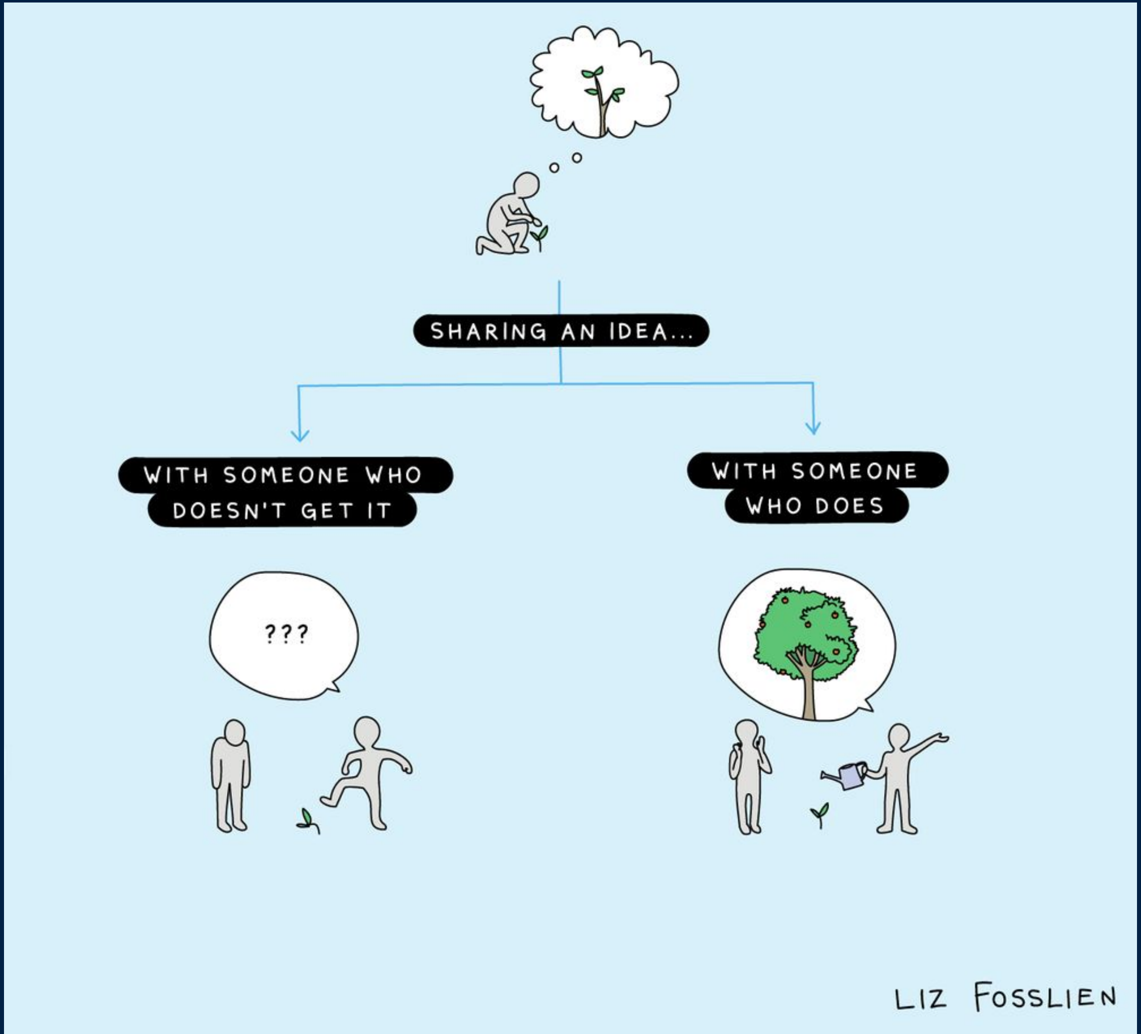


Wrap up


SHARING AN IDEA...
WITH SOMEONE WHO DOESN'T GET IT
???
WITH SOMEONE WHO DOES
LIZ FOSSLIEN